

AI-Native Homebuilder Architecture / Integration

The agent-based operating system for home builders — mapped to the MTP + DRIVE + SHAPE framework

The thesis: Every other construction platform is software your team logs into. Cornerstone is software that logs in *for* your team. The legal entity stays the accountability shell — purpose, liability, IP, brand, judgment. Everything else moves into a governed agent stack that senses, interprets, decides, orchestrates, and learns.

ARCHITECTURE LAYER	WHAT IT MEANS	CORNERSTONE IMPLEMENTATION LIVE
Accountability Shell	The legal entity holds purpose, liability, IP, brand, customer trust, compliance, capital, and human judgment.	Your builder remains the licensed, insured, signing party. Cornerstone is the OS underneath — never the customer-of-record.
Purpose Protocol (MTP)	Mission encoded as a machine-readable operating constraint — on-time, on-budget, code-compliant, customer trust.	Foreman AI system prompt + 39 estimating formulas + 10 prompt categories + policy boundaries. Purpose is enforced, not framed on a wall.
Sensing	Agents monitor signals — schedule slips, vendor delays, customer messages, financials, competitors.	30 named webhook events with HMAC-signed payloads. Foreman monitors bid deadlines, schedule cascades, customer comms, vendor responsiveness.
Interpretation	Agents explain what signals mean — bid comparisons, risk scoring, variance analysis.	Side-by-side bid comparison, vendor scorecards, profitability reports, budget vs actual reports, schedule cascade analysis.
Decision	Agents propose options or decisions.	Foreman proposes option swaps, vendor selections, schedule resequences, SOW edits, punch-list priorities. Builder approves.
Orchestration	Agents assign work, trigger workflows, route tasks, coordinate other agents and humans.	396+ Foreman skills across 20 categories. Reads AND writes the database. Sends vendor notifications, updates design options, generates bid templates, creates purchase orders.
Learning	Agents review outcomes and improve the process. The new moat.	Per-user memory + company memory. Foreman remembers preferences, vendors, communication style. Built-in context compaction — works through 200+ option cleanups without forgetting.
Govern & Assure	Audit logs, approval gates, rollback, permission controls, human review queues.	Per-action audit trail. Approval gates on bids, change orders, customer-facing comms. Role-based permissions. CSV/JSON full-DB export anytime. Nightly auto-backups.

The Digital Twin Pilot

- **Pick one workflow** — bid → award → community assignment is highest-leverage
- Fork your vendor + scope data into a parallel Cornerstone instance
- Run AI-native version alongside current process for 60 days
- Measure: response time, error rate, PM hours saved, awarded-price accuracy
- Expand to next workflow only after the twin wins on every metric

Why Cornerstone, Not a Generic AI

- 396 construction-specific skills, not 5 generic chat tools
- 183KB / 24,500-word domain knowledge base built in
- Community-assigned vendor wins — respects long-term sub relationships
- Lock-after-acceptance protects awarded pricing from drift
- BYOA: Pro+ tier exposes 150+ REST endpoints + 30 webhook events
- ~\$0.01 CRM migration (Buildertrend, JobTread, CoConstruct, CSV) vs. 5-figure white-glove fees

The Intelligence Moat: The old moat was headcount, capital, and distribution. The new moat is proprietary data + proprietary workflows + fast learning loops. Buildertrend has 1M+ users and zero learning — their default schedule durations in 2026 are the same as 2018. Foreman gets smarter every job. The longer you use Cornerstone, the harder it is to leave — not because of lock-in, but because it knows your business.

Reference Agent Architecture for a Home Builder

What the AI-native operating system actually looks like — agent count, roles, and the platforms underneath

AGENT STACK — AT A GLANCE

5
TIERS

12
AGENTS

396+
SKILLS

30
WEBHOOKS

APPROVAL GATES & GOVERNANCE

150+
REST APIS

100%
AUDITED

JSON
EXPORTABLE

RBAC
PER-ROLE

TIER / AGENT	RESPONSIBILITY	PLATFORM / TECH STACK
Builder Principal TIER 0 · HUMAN	Sets strategy and MTP. Owns liability, brand, capital. Approves bid awards over threshold, customer commitments, contract changes, hiring.	Human owner / executive. Authority is non-delegable.
Orchestrator TIER 1 · ROUTING	Routes inbound work (email, SMS, voice, in-app) to the right specialist agent. Maintains conversation context across handoffs.	Cornerstone Foreman AI · Anthropic Claude Sonnet / OpenAI GPT-5 · LangGraph + CrewAI orchestration · Postgres state
Schedule Agent TIER 2 · DECISION	Monitors task slippage, cascades downstream dates, flags critical-path risk, proposes resequences.	Foreman skills (32 scheduling) · Gantt engine · cron triggers · Slack/SMS escalation
Vendor / Bid Agent TIER 2 · DECISION	Generates per-trade bid templates, compares submissions, scorecards vendors, recommends award by community.	Foreman skills · Excel template generator · Vercel Blob storage · HMAC vendor portal
Selections Agent TIER 2 · DECISION	Tracks buyer selection deadlines, drafts reminders, locks Designer Packages, prices options against real vendor data.	Foreman skills (49 Design Center + 23 Option Attributes) · vision LLM for spec sheets
Customer Agent TIER 2 · DECISION	Drafts weekly buyer updates, handles common Q&A, escalates anything sentiment-flagged or contract-impacting.	Cornerstone Postman · nodemailer · Twilio SMS · BlueBubbles iMessage · Customer Portal
Finance / Margin Agent TIER 2 · DECISION	Reviews invoices vs POs, flags margin erosion, runs profitability + budget-vs-actual reports per home and community.	Foreman skills · QuickBooks / Sage connectors (roadmap) · Master Cost Budget engine
Permitting / Compliance Agent TIER 2 · DECISION	Tracks permit submittals, inspection results, jurisdiction-specific code requirements. Escalates failed inspections.	Foreman skills · Permitting Pipeline UI · jurisdiction rule packs · document OCR
Voice / SMS Agents TIER 3 · EXECUTION	AI voice + SMS agents that call/text vendors, buyers, and field crews. Wired to Cornerstone webhooks for bid follow-ups, jobsite check-ins, selection reminders, delivery confirmations.	Retell AI (voice + SMS) · Bland.ai (outbound voice) · Twilio (SMS infra) · BlueBubbles (iMessage) · webhook-triggered
BYOA Hooks TIER 3 · EXECUTION	Customer-owned agents wired to Cornerstone for custom workflows beyond the built-in voice stack — internal tools, accounting, MLS, third-party CRMs.	30 webhook events (HMAC-signed) · 150+ REST endpoints · Make · n8n · Zapier · custom integrations
Migration Agent TIER 3 · EXECUTION	One-click import from existing builder software. AI auto-detects source platform and column mappings.	CRM Migration Wizard · LLM column mapper · supports Buildertrend / JobTread / CoConstruct / Procore / CSV (~\$0.01/migration)
Learning Agent TIER 4 · MEMORY	Persists per-user + company memory. Tunes vendor scores, schedule durations, estimating formulas from outcomes over time.	Foreman memory store · context compaction · 39 estimating formulas · vector embeddings (pgvector)
Govern & Assure TIER 4 · CONTROL	Audit trail on every write. Approval gates. Role-based permissions. Full DB export anytime. Nightly backups + on-demand snapshots.	Postgres audit tables · Next.js RBAC middleware · CSV/JSON export · webhook delivery logs · SOC2-aligned controls

Foundation Platforms

- **Next.js 15** App Router
- **Postgres** (Neon serverless)
- **Vercel** edge + serverless
- **Vercel Blob** file storage
- **Prisma** ORM + migrations

AI / Agent Layer

- **Anthropic Claude** Sonnet / Opus
- **OpenAI GPT-5** + vision
- **LangGraph** orchestration
- **CrewAI** multi-agent runtime
- **Custom skill registry** (396+)
- **pgvector** embeddings

Voice / SMS Stack

- **Retell AI** voice + SMS
- **Bland.ai** outbound voice
- **Twilio** SMS + phone
- **BlueBubbles** iMessage

Integration Surface

- **REST API** 150+ endpoints
- **Webhooks** 30 events, HMAC
- **Vendor Portal** token auth
- **Customer Portal** branded

How to read this: Tiers 0–1 are non-negotiable. Tiers 2–3 are modular — start with the workflow (digital twin) that hurts most and add agents as the twin proves out. Tier 4 (governance + learning) is the backbone — what separates a real business OS from a chatbot pilot.

AI Voice & SMS Agent Roster

Built on Retell AI, Bland.ai, Twilio, and BlueBubbles — every voice/SMS touchpoint logged and approval-gated

Vendor Bid Outreach

OUTBOUND

Retell AI voice · fallback to Twilio SMS

Calls invited subs to confirm receipt of bid invitation, walks them through the Excel template, answers scope questions, captures verbal commitment to deadline.

trigger: webhook bid_request.sent + 24h no-view

Bid Deadline Reminder

OUTBOUND

Retell AI voice · Twilio SMS escalation

T-48h and T-12h reminders to vendors who haven't submitted. Offers extension request, captures reason if declining, escalates to PM if vendor unreachable.

trigger: cron + webhook bid_request.deadline_approaching

Bid Award Notification

OUTBOUND

Retell AI voice · Postman email follow-up

Calls awarded vendor to confirm acceptance, schedules kickoff, sends contract via DocuSign. Calls non-winners with respectful decline + community-fit feedback.

trigger: webhook bid.awarded + bid.declined

Vendor Inbound Hotline

INBOUND

Retell AI · vendor-portal token auth via phone DTMF

Vendors call one number to check bid status, request extensions, ask scope questions, or update availability. Authenticates by phone number lookup.

trigger: inbound call to Twilio number

Jobsite Check-In

2-WAY SMS

Retell SMS · Twilio fallback

Texts vendors morning-of for jobsite confirmation, ETA, crew size. Replies parsed into Foreman schedule. Photo replies attached to home record.

trigger: scheduled task day-of · webhook task.starts_today

Buyer Selection Nudge

2-WAY SMS

Retell SMS · BlueBubbles iMessage relay

Texts buyers about overdue design selections, links back to Customer Portal, answers basic questions. Escalates contractual issues to sales agent.

trigger: webhook selection.deadline_missed

Delivery Confirmation

OUTBOUND

Bland.ai (high-volume outbound) · Twilio SMS

Calls suppliers day-before material delivery to confirm SKU, quantity, jobsite gate code, and dropoff window. Updates PO status on confirmation.

trigger: webhook po.delivery_tomorrow

Buyer Inbound Concierge

INBOUND

Retell AI voice · routes to Foreman context

Answers buyer questions 24/7 about home status, selection deadlines, walkthrough scheduling. Has read-only Foreman context. Hands off to humans for contract or warranty issues.

trigger: inbound call to community line

End-to-End Scenario: Vendor Bid Workflow with Voice Agents

DAY 0

VENDOR/BID Foreman generates per-trade Excel templates for 8 subs across 3 communities. Sends email + portal links. Fires bid_request.sent webhook.

+24h

RETELL VOICE Vendor Bid Outreach calls the 3 subs who haven't opened the portal. Confirms scope, walks them through the template, captures verbal yes/no.

+5 days

RETELL SMS Bid Deadline Reminder texts the 2 subs still pending at T-48h. One requests extension via text — parsed by AI, surfaced to PM for approval.

+7 days

FOREMAN All 8 bids submitted. Side-by-side comparison auto-generated. Vendor scorecard pulled from past job history. PM reviews + awards.

+7 days

RETELL VOICE Bid Award Notification calls winning vendor (verbal confirm + kickoff scheduled), then calls 7 non-winners with respectful decline + per-community feedback.

+8 days

POSTMAN Contract auto-generated, DocuSign sent. Awarded price locked. Webhook bid.awarded fires to any BYOA hooks for accounting/CRM sync.

Day-of

RETELL SMS Jobsite Check-In texts vendor morning of start. Vendor confirms ETA + crew. Reply parsed, schedule updated, PM notified.

What this replaces: 3-5 PM hours per bid cycle on follow-up calls, voicemails, status texts, contract chasing, and reminder emails. Multiply by 8 subs × 3 communities × 4 cycles/yr — that's ~400+ PM hours/yr per builder, just on bid coordination. Every call is logged, transcribed, and human-reviewable.

Memory Architecture — The Company Brain

How Cornerstone stores, retrieves, and governs agent memory at the database layer — not in loose markdown files

The problem with file-based memory: Loose .md files have no permissions, audit, dedup, expiration, search, or rollback. **Cornerstone's memory lives in Postgres + pgvector with markdown as a compiled fast-bootstrap view on top.** Every write is validated, scored, deduplicated, audit-logged, and reversible.

The 4-Layer Memory Stack



Memory Write Pipeline



Core Memory Tables (Neon Postgres)

agent_memories

```
CREATE TABLE agent_memories (  
  id          uuid PRIMARY KEY,  
  agent_name  text NOT NULL,  
  memory_type text NOT NULL, -- fact|rule|preference|event  
  subject     text,  
  content     text NOT NULL,  
  importance  int DEFAULT 3,  -- 1-5 scale  
  confidence  numeric(3,2),  
  source      text,          -- which conversation/event  
  embedding   vector(1536),  -- pgvector  
  expires_at  timestampz,  
  created_at  timestampz DEFAULT now()  
);
```

Per-user + company-wide memories. Foreman writes here every time it learns a builder preference, vendor pattern, or workflow rule.

memory_audit

```
CREATE TABLE memory_audit (  
  id          uuid PRIMARY KEY,  
  memory_id   uuid REFERENCES agent_memories,  
  action      text,          -- insert|update|delete  
  proposed_by text,          -- which agent  
  approved_by text,          -- human or orchestrator  
  previous    jsonb,         -- full prior state  
  new_value   jsonb,  
  reason      text,  
  created_at  timestampz DEFAULT now()  
);
```

Every memory change. Full rollback by replaying previous. SOC2-friendly audit trail. Searchable by agent, time range, or affected memory.

workflow_rules

```
CREATE TABLE workflow_rules (  
  id          uuid PRIMARY KEY,  
  scope       text,          -- company|community|user  
  trigger_event text,        -- webhook name  
  condition   jsonb,  
  action      jsonb,         -- skill + params  
  approval_req boolean,  
  active      boolean DEFAULT true,  
  created_by  text  
);
```

Builder-defined automation rules. E.g. "when bid lockout fires, call vendor on Retell at +24h." Lance can author and version these.

vendor_scorecards

```
CREATE TABLE vendor_scorecards (  
  vendor_id      uuid REFERENCES vendors,  
  community_id   uuid REFERENCES communities,  
  bid_accuracy   numeric(4,2), -- bid vs actual  
  on_time_pct    numeric(4,2),  
  rework_rate    numeric(4,2),  
  responsiveness numeric(4,2), -- avg reply hrs  
  total_jobs     int,  
  last_updated   timestampz,  
  PRIMARY KEY (vendor_id, community_id)  
);
```

The learning loop in action. Updated nightly from job outcomes. Drives vendor recommendation in Tier 2 Vendor/Bid agent.

Other Tables in the Brain

Entities

- companies
- users + roles
- vendors
- contacts
- communities
- homes

Knowledge

- knowledge_docs
- estimating_formulas
- prompt_library
- decisions
- preferences

Session

- conversations
- session_state
- daily_logs
- tool_calls
- tasks + events

Governance

- memory_audit
- webhook_deliveries
- approval_queue
- permissions
- backups

Why This Matters for a Builder

- Per-tenant isolation:** Your data lives in its own Postgres schema with row-level security — not in a shared table with other builders.
- Searchable institutional memory:** Foreman remembers which floorplans + communities get which subs, which vendors bid high but deliver, which buyers always push selection deadlines.
- Learning compounds:** Every job teaches the system. Bid accuracy, schedule durations, vendor responsiveness all tune from real outcomes — not vendor marketing claims.
- Audit on demand:** "Show every memory write Foreman made about Vendor X in Q1" — one SQL query, fully reproducible, defensible in disputes.
- Your data leaves with you:** Full Postgres dump anytime. Schema documented. Migration is portable, not hostage.